QA System - LLM-Based Evaluation 0.1.0

OAS 3.1

/openapi.json

Automated QA Generator with LLM-Based Evaluation

Metrics:

- **Faithfulness (Ragas)**: LLM verifies answer grounding
- **Answer Relevancy (Ragas)**: LLM + embeddings measure relevance

Endpoints:

- POST /ask: Upload file + ask question
- POST /generate: Generate Q&A pairs from document

Fully LLM-based evaluation satisfies project requirements!

default

POST /ask Ask

Parameters Cancel Reset

No parameters

Request body required multipart/form-data

file * required
string($binary)
Choose File semi.txt

question * required
string
what is ai?

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@semi.txt;type=text/plain' \
  -F 'question=what is ai?'
```

Request URL

```
http://127.0.0.1:8000/ask
```

Server response

Code	Details
200	**Response body**

```
{
  "question": "what is ai?",
  "answer": "This information is not available in the uploaded document.",
  "faithfulness_score": null,
  "relevancy_score": null,
  "combined_score": null,
  "evaluation_summary": "N/A — Information not found in document"
  "evaluation_method": "LLM-based (Ragas with qwen/qwen3-32b)"
}
```

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Response headers

```
content-length: 293
content-type: application/json
date: Thu,19 Mar 2026 14:41:00 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema

```
{
  "question": "string",
  "answer": "string",
```

Code	Description	Links


```
  "faithfulness_score": 0,
  "relevancy_score": 0,
  "combined_score": 0,
  "evaluation_summary": "string",
  "evaluation_method": "LLM-based (Ragas)"
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


POST `/generate` Generate

Parameters Cancel Reset

No parameters

Request body required **multipart/form-data**

file * required
`string($binary)` Choose File semi.txt

num_pairs
`integer` 5

Send empty value

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/generate' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@semi.txt;type=text/plain' \
  -F 'num_pairs=5'
```

Request URL

```
http://127.0.0.1:8000/generate
```

Server response

Code	Details
200	

Response body


```
{
  "total_pairs": 5,
  "average_faithfulness": 1,
  "average_relevancy": 0.59,
  "average_combined": 0.8,
  "evaluation_method": "LLM-based (Ragas with qwen/qwen3-32b)",
  "pairs": [
    {
      "question": "What is the lower energy band where electrons are tightly bou
nd to atoms and cannot move freely?",
      "answer": "The Valence Band is the lower energy band where electrons are t
ightly bound to atoms and cannot move freely.",
      "faithfulness_score": 1,
      "relevancy_score": 0.46,
      "combined_score": 0.73
    },
    {
      "question": "What is the energy difference between the valence band and th
e conduction band?",
      "answer": "The Forbidden Energy Gap is the energy difference between the v
alence band and the conduction band.",
      "faithfulness_score": 1,
      "relevancy_score": 0.41,
      "combined_score": 0.7
    },
    {
      "question": "How do semiconductors become N-type?",
      "answer": "N-type semiconductors are created by adding pentavalent impurit
```


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Response headers

```
content-length: 1505
content-type: application/json
date: Thu,19 Mar 2026 14:37:13 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Code	Description	Links

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "total_pairs": 0,
  "average_faithfulness": 0,
  "average_relevancy": 0,
  "average_combined": 0,
  "evaluation_method": "LLM-based (Ragas)",
  "pairs": [
    {
      "question": "string",
      "answer": "string",
      "faithfulness_score": 0,
      "relevancy_score": 0,
      "combined_score": 0
    }
  ]
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


Schemas

AskResponse > Expand all `object`

Body_ask_ask_post > Expand all `object`

Body_generate_generate_post > Expand all `object`

GenerateResponse > Expand all `object`

HTTPValidationError > Expand all `object`

QAPair > Expand all `object`

ValidationError > Expand all `object`